

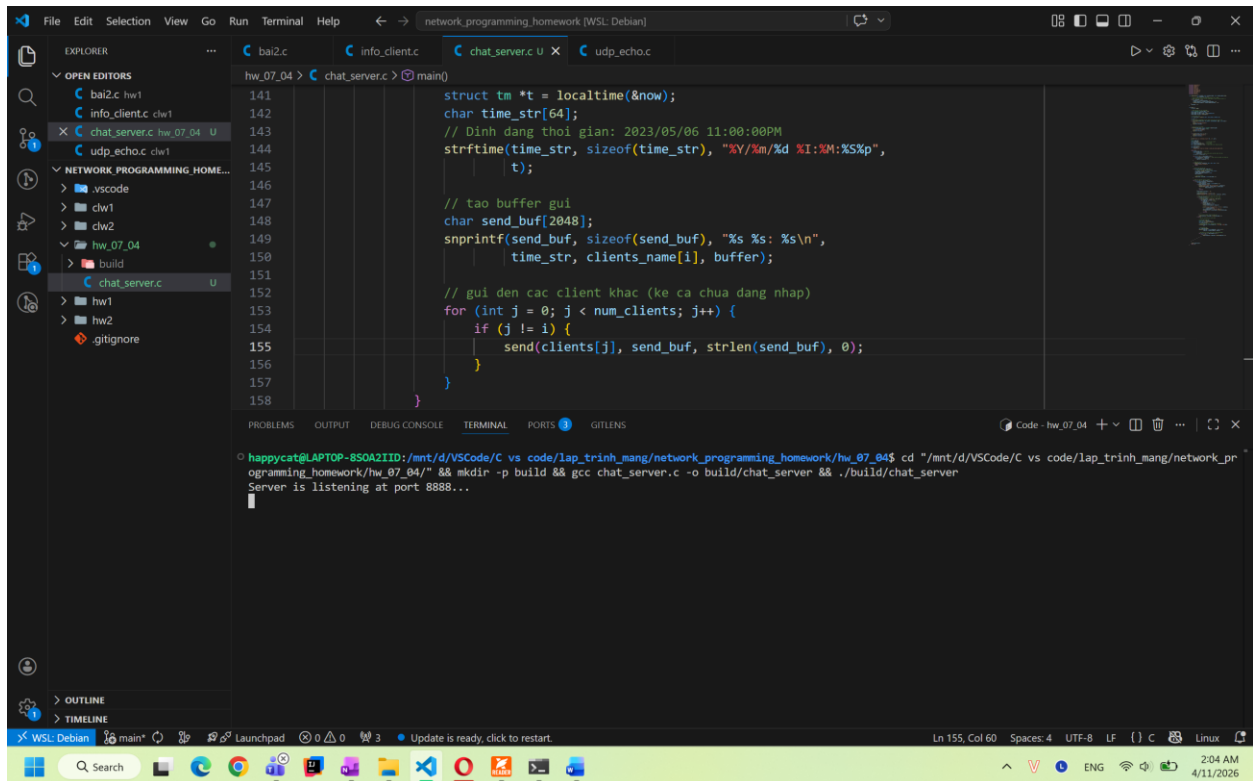
BÁO CÁO BÀI TẬP VỀ NHÀ SỐ 2 (NGÀY 07/04/2026)

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1. Bài 1 – chat_server

- Chat server ở cổng 8888



The screenshot displays a Visual Studio Code (VS Code) environment with a C program for a chat server. The Explorer sidebar on the left shows the project structure, including files like `bai2.c`, `info_client.c`, `chat_server.c`, and `udp_echo.c`. The main editor window shows the `chat_server.c` file, which contains the following code:

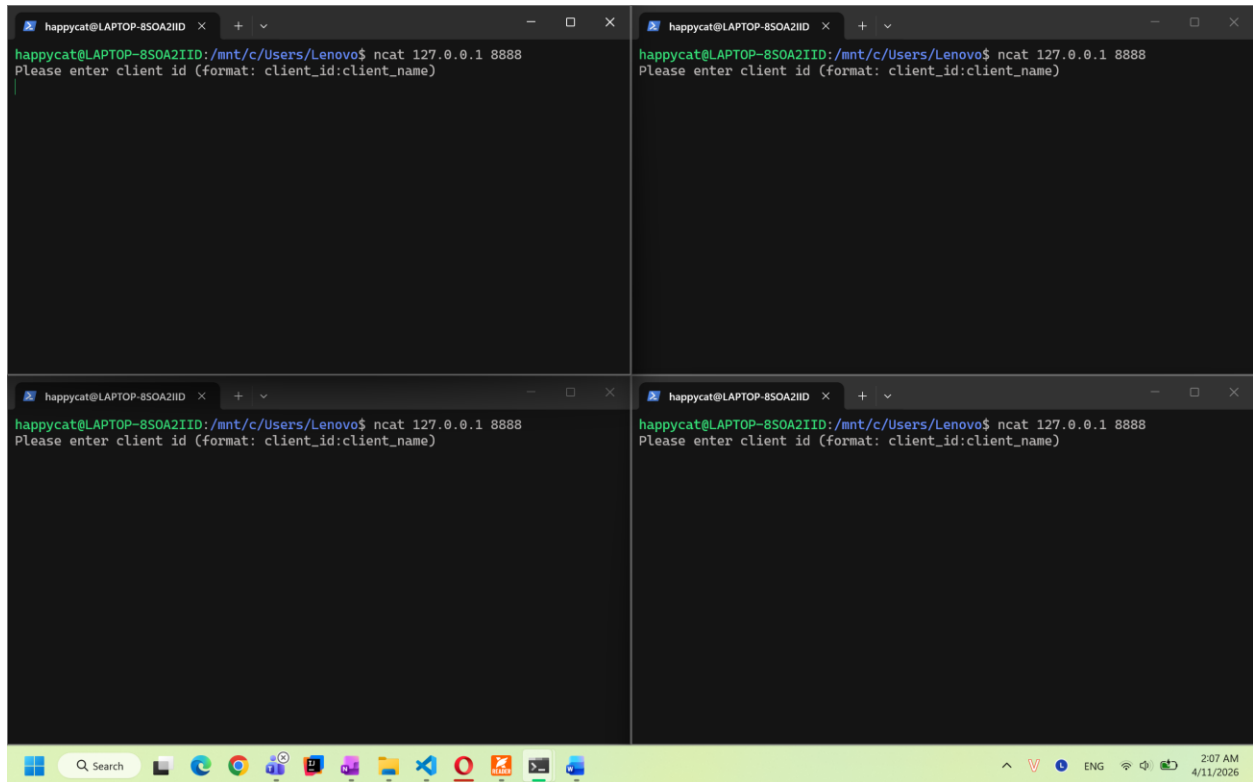
```
141 struct tm *t = localtime(&now);
142 char time_str[64];
143 // Định dạng thời gian: 2023/05/06 11:00:00PM
144 strftime(time_str, sizeof(time_str), "%Y/%m/%d %I:%M:%S%p",
145          t);
146
147 // tạo buffer gửi
148 char send_buf[2048];
149 snprintf(send_buf, sizeof(send_buf), "%s %s: %s\n",
150          time_str, clients_name[i], buffer);
151
152 // gửi đến các client khác (kể cả chưa đăng nhập)
153 for (int j = 0; j < num_clients; j++) {
154     if (j != i) {
155         send(clients[j], send_buf, strlen(send_buf), 0);
156     }
157 }
158 }
```

The terminal window at the bottom shows the command used to compile and run the program:

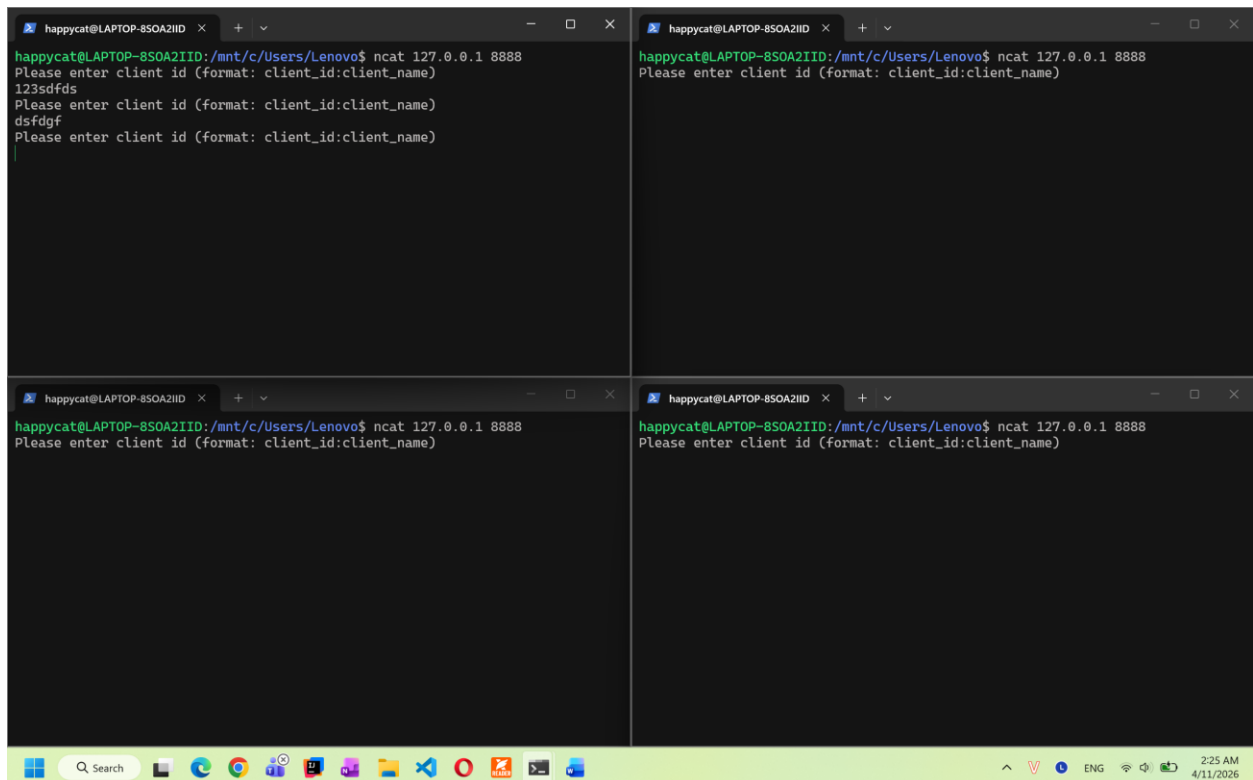
```
happyat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_07_04$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_07_04/" && mkdir -p build && gcc chat_server.c -o build/chat_server && ./build/chat_server
Server is listening at port 8888...
```

The status bar at the bottom indicates the file is at line 155, column 60, with 4 spaces, UTF-8 encoding, and LF line endings. The system clock shows 2:04 AM on 4/11/2026.

- Khi client kết nối, server gửi yêu cầu nhập client id



- Nếu client nhập sai, server sẽ gửi lại yêu cầu



- Client nhập đúng cú pháp thì server sẽ đánh dấu là đã nhập và gán tên cho client

```
happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
123sdfds
Please enter client id (format: client_id:client_name)
dsfdgf
Please enter client id (format: client_id:client_name)
client_id:cl1
Bat dau chat:

happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl2
Bat dau chat:

happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl3
Bat dau chat:

happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl4
Bat dau chat:
```

- Client 1 gửi thì các client còn lại sẽ nhận được dữ liệu và thời gian

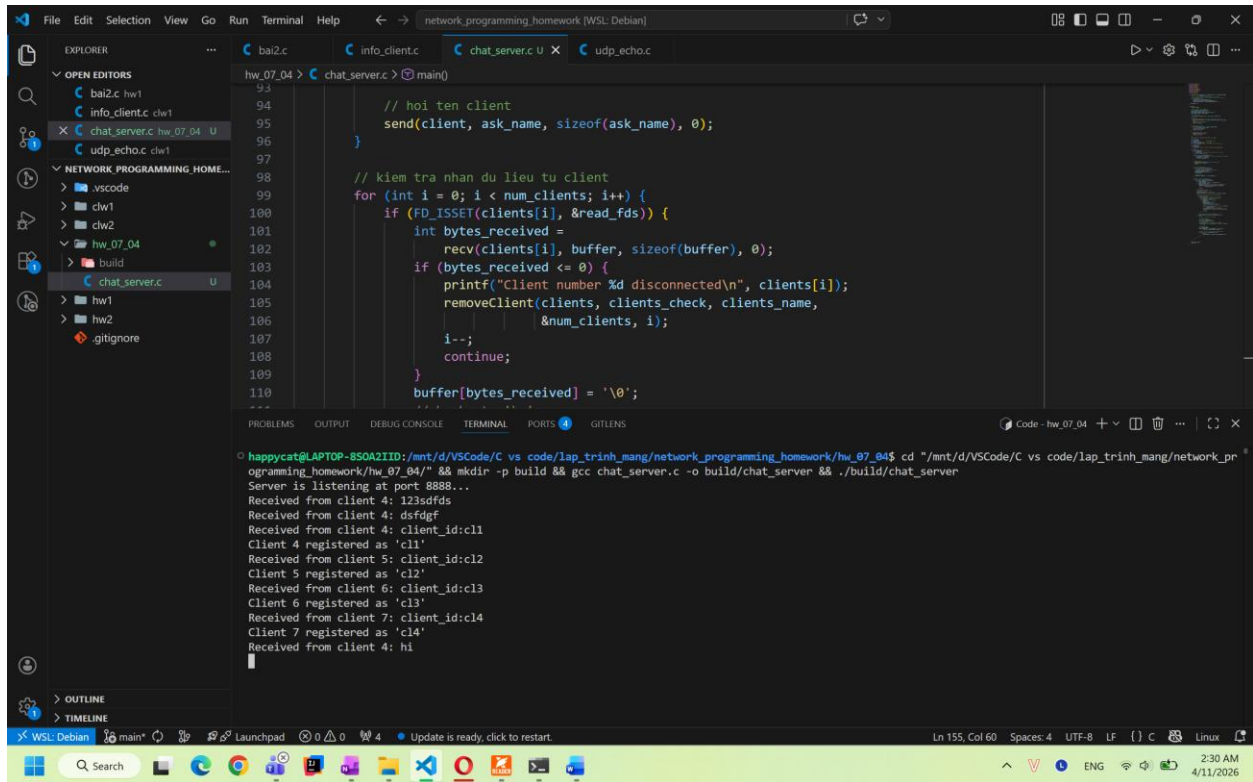
```
happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
123sdfds
Please enter client id (format: client_id:client_name)
dsfdgf
Please enter client id (format: client_id:client_name)
client_id:cl1
Bat dau chat:
hi

happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl2
Bat dau chat:
2026/04/11 02:30:20AM cl1: hi

happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl3
Bat dau chat:
2026/04/11 02:30:20AM cl1: hi

happycat@LAPTOP-85OA2IID /mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:cl4
Bat dau chat:
2026/04/11 02:30:20AM cl1: hi
```

- Kết quả phía server



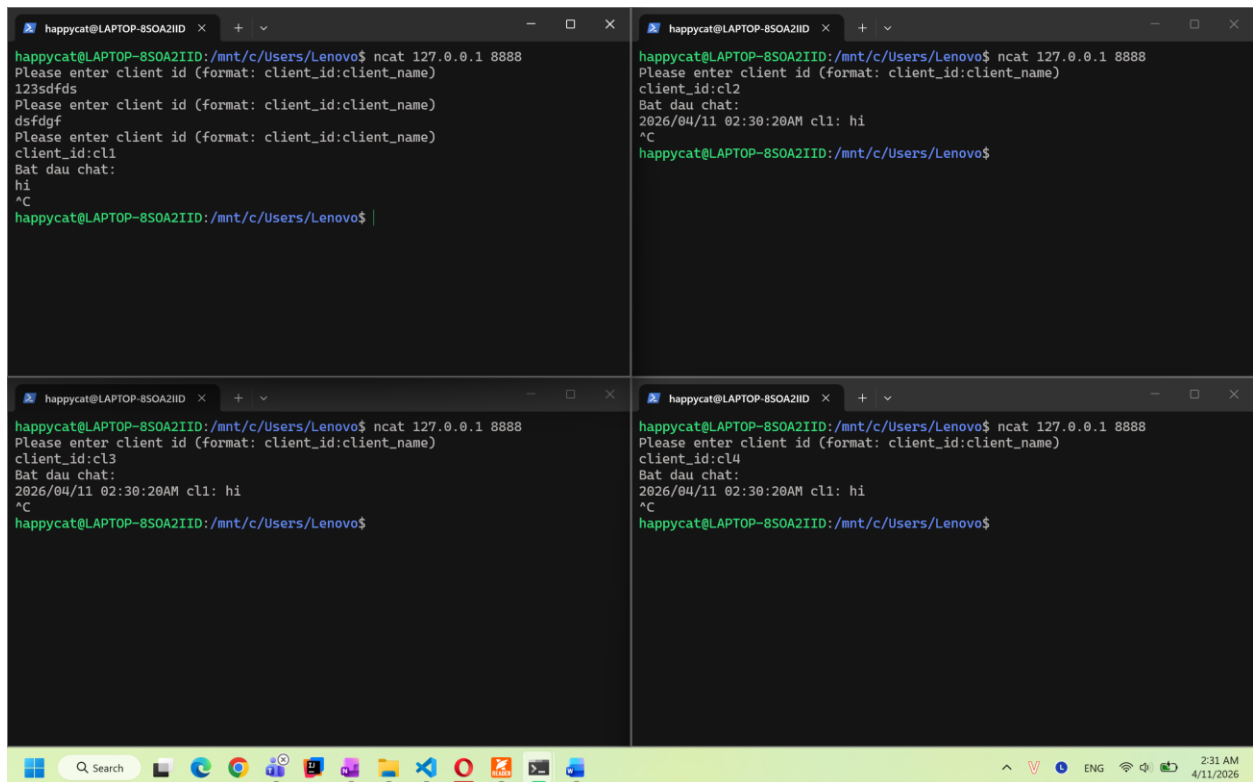
The screenshot shows the VS Code editor with the `chat_server.c` file open. The code implements a chat server that listens on port 8888, receives client connections, and processes messages. The terminal output shows the server's execution, including client registration and message handling.

```
hw_07_04 > chat_server.c > main()
93
94 // hoi ten client
95 send(client, ask_name, sizeof(ask_name), 0);
96
97
98 // Kiem tra nhan du lieu tu client
99 for (int i = 0; i < num_clients; i++) {
100     if (FD_ISSET(clients[i], &read_fds)) {
101         int bytes_received =
102             recv(clients[i], buffer, sizeof(buffer), 0);
103         if (bytes_received <= 0) {
104             printf("Client number %d disconnected\n", clients[i]);
105             removeClient(clients, clients_check, clients_name,
106                           &num_clients, i);
107             i--;
108             continue;
109         }
110         buffer[bytes_received] = '\0';
```

Terminal Output:

```
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_07_04$ cd ~/mnt/d/VSCode/C vs code/lap_trinh_mang/network_pr
ogramming_homework/hw_07_04/" && mkdir -p build && gcc chat_server.c -o build/chat_server && ./build/chat_server
Server is listening at port 8888...
Received from client 4: 123sdfds
Received from client 4: dsfdgf
Received from client 4: client_id:c11
Client 4 registered as 'cl1'
Received from client 5: client_id:c12
Client 5 registered as 'cl2'
Received from client 6: client_id:c13
Client 6 registered as 'cl3'
Received from client 7: client_id:c14
Client 7 registered as 'cl4'
Received from client 4: hi
```

- Khi client ngắt kết nối



The screenshot shows four terminal windows, each representing a client's interaction with the chat server. The clients enter their IDs and names, and the server responds with a confirmation message. The clients then press Ctrl+C to disconnect, and the server responds with a message indicating the client has disconnected.

```
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
123sdfds
Please enter client id (format: client_id:client_name)
dsfdgf
Please enter client id (format: client_id:client_name)
client_id:c11
Bat dau chat:
hi
^C
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$
```

```
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:c12
Bat dau chat:
2026/04/11 02:30:20AM cl1: hi
^C
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$
```

```
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:c13
Bat dau chat:
2026/04/11 02:30:20AM cl1: hi
^C
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$
```

```
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter client id (format: client_id:client_name)
client_id:c14
Bat dau chat:
2026/04/11 02:30:20AM cl1: hi
^C
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$
```

```
File Edit Selection View Go Run Terminal Help network_programming_homework [WSL: Debian]

EXPLORER
  bai2.c
  info_client.c
  chat_server.c
  udp_echo.c

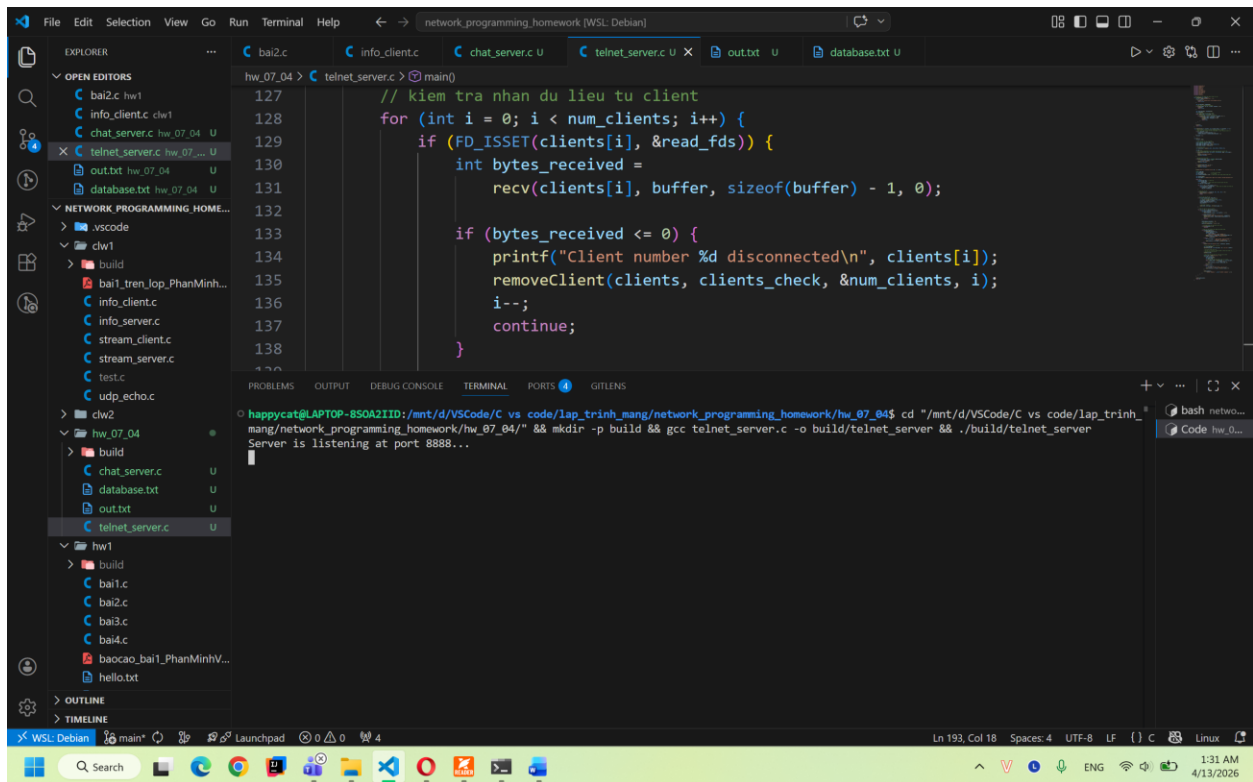
OPEN EDITORS
  bai2.c
  info_client.c
  chat_server.c
  udp_echo.c

NETWORK PROGRAMMING HOMEWORK
  .vscode
  clw1
  clw2
  hw_07_04
  build
  chat_server.c
  hw1
  hw2
  .gitignore

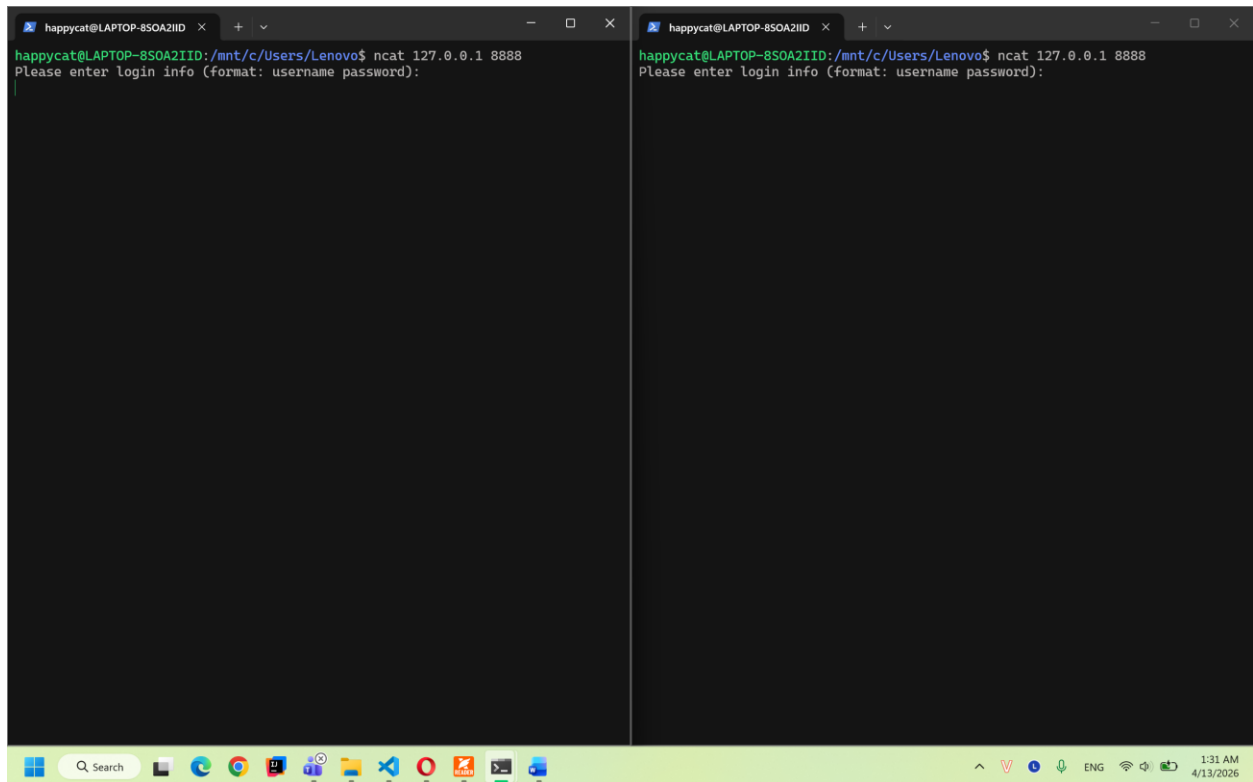
chat_server.c
  93 // hoi ten client
  94 send(client, ask_name, sizeof(ask_name), 0);
  95 }
  96
  97 // Kiem tra nhan du lieu tu client
  98 for (int i = 0; i < num_clients; i++) {
  99     if (FD_ISSET(clients[i], &read_fds)) {
 100         int bytes_received =
 101             recv(clients[i], buffer, sizeof(buffer), 0);
 102         if (bytes_received <= 0) {
 103             printf("Client number %d disconnected\n", clients[i]);
 104             removeClient(clients, clients_check, clients_name,
 105                         &num_clients, i);
 106             i--;
 107             continue;
 108         }
 109         buffer[bytes_received] = '\0';
 110     }
 111 }
```

```
happy@LAPTOP-850A2IID: /mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_07_04$ cd ~/mnt/d/VSCode/C vs code/lap_trinh_mang/network_pr
ogramming_homework/hw_07_04$ && mkdir -p build && gcc chat_server.c -o build/chat_server && ./build/chat_server
Server is listening at port 8888...
Received from client 4: 12345678
Received from client 4: dsfdgf
Received from client 4: client_id:c11
Client 4 registered as 'c11'
Received from client 5: client_id:c12
Client 5 registered as 'c12'
Received from client 6: client_id:c13
Client 6 registered as 'c13'
Received from client 7: client_id:c14
Client 7 registered as 'c14'
Received from client 4: hi
Client number 4 disconnected
Client number 5 disconnected
Client number 6 disconnected
Client number 7 disconnected
```

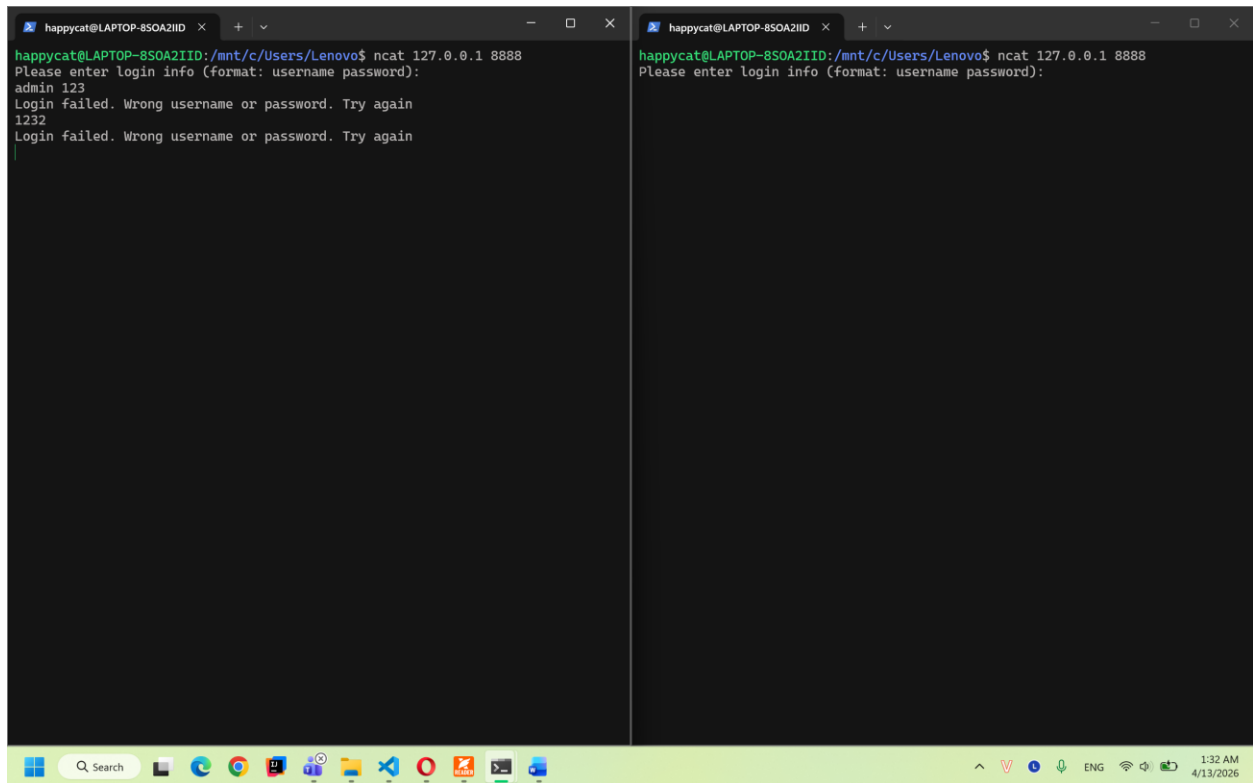
2. Bài 2 – telnet_server
- Chạy server ở cổng 8888



- Chạy 2 client kết nối vào server

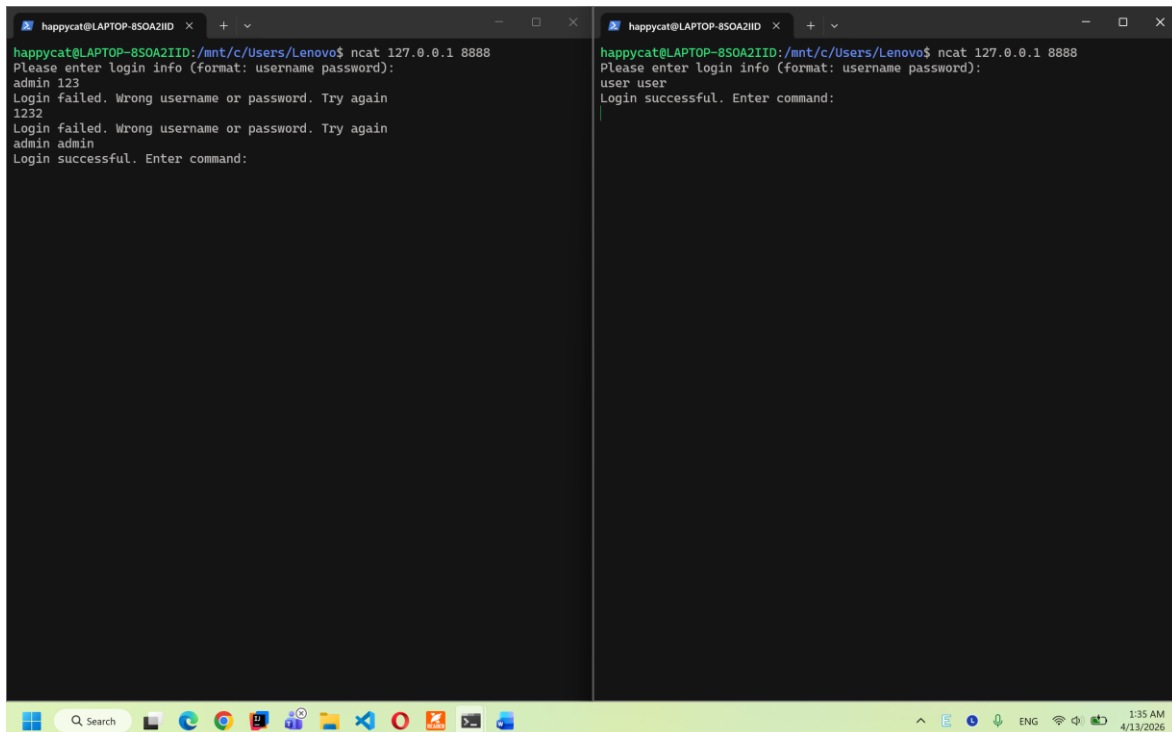


- Nhập username và password (format: “username password”)
- Nếu nhập sai, server sẽ báo lỗi và yêu cầu nhập lại



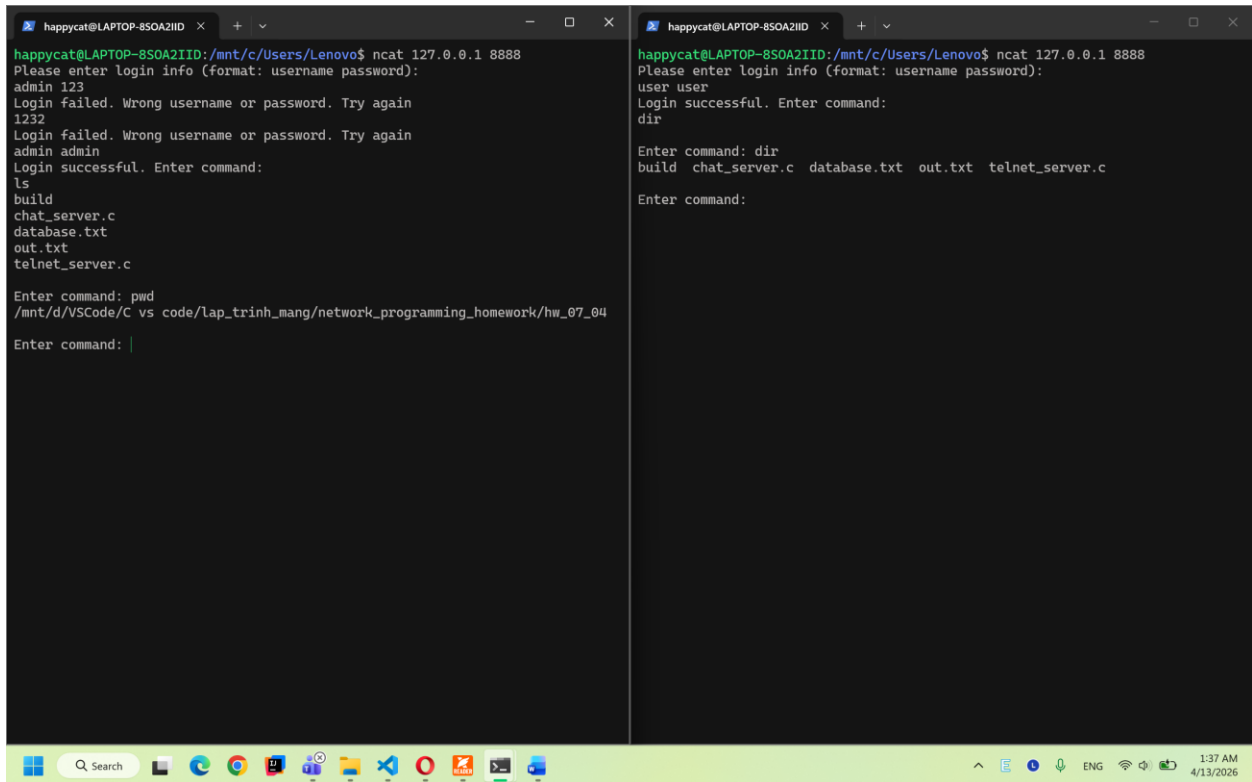
The image shows two side-by-side terminal windows. Both windows have a title bar that reads 'happycat@LAPTOP-850A2IID'. The left terminal window shows the following text: 'happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo\$ ncat 127.0.0.1 8888', 'Please enter login info (format: username password):', 'admin 123', 'Login failed. Wrong username or password. Try again', '1232', 'Login failed. Wrong username or password. Try again'. The right terminal window shows the same initial text: 'happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo\$ ncat 127.0.0.1 8888', 'Please enter login info (format: username password):'. The Windows taskbar is visible at the bottom, showing the search bar, task view, and various application icons. The system clock in the bottom right corner indicates 1:32 AM on 4/13/2026.

- Nhập đúng thì server sẽ báo thành công và yêu cầu nhập lệnh



The image shows two side-by-side terminal windows. Both windows have a title bar that reads 'happycat@LAPTOP-850A2IID'. The left terminal window shows the following text: 'happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo\$ ncat 127.0.0.1 8888', 'Please enter login info (format: username password):', 'admin 123', 'Login failed. Wrong username or password. Try again', '1232', 'Login failed. Wrong username or password. Try again', 'admin admin', 'Login successful. Enter command:'. The right terminal window shows the same initial text: 'happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo\$ ncat 127.0.0.1 8888', 'Please enter login info (format: username password):', followed by 'user user' and 'Login successful. Enter command:'. The Windows taskbar is visible at the bottom, showing the search bar, task view, and various application icons. The system clock in the bottom right corner indicates 1:35 AM on 4/13/2026.

- Nhập lệnh xong, server sẽ thực hiện, ghi kết quả ra file out.txt và gửi về client



The image shows two terminal windows side-by-side. The left window is the server terminal, and the right window is the client terminal. Both are running on a Windows machine with WSL2.

Server Terminal:

```
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter login info (format: username password):
admin 123
Login failed. Wrong username or password. Try again
1232
Login failed. Wrong username or password. Try again
admin admin
Login successful. Enter command:
ls
build
chat_server.c
database.txt
out.txt
telnet_server.c

Enter command: pwd
/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_07_04

Enter command: |
```

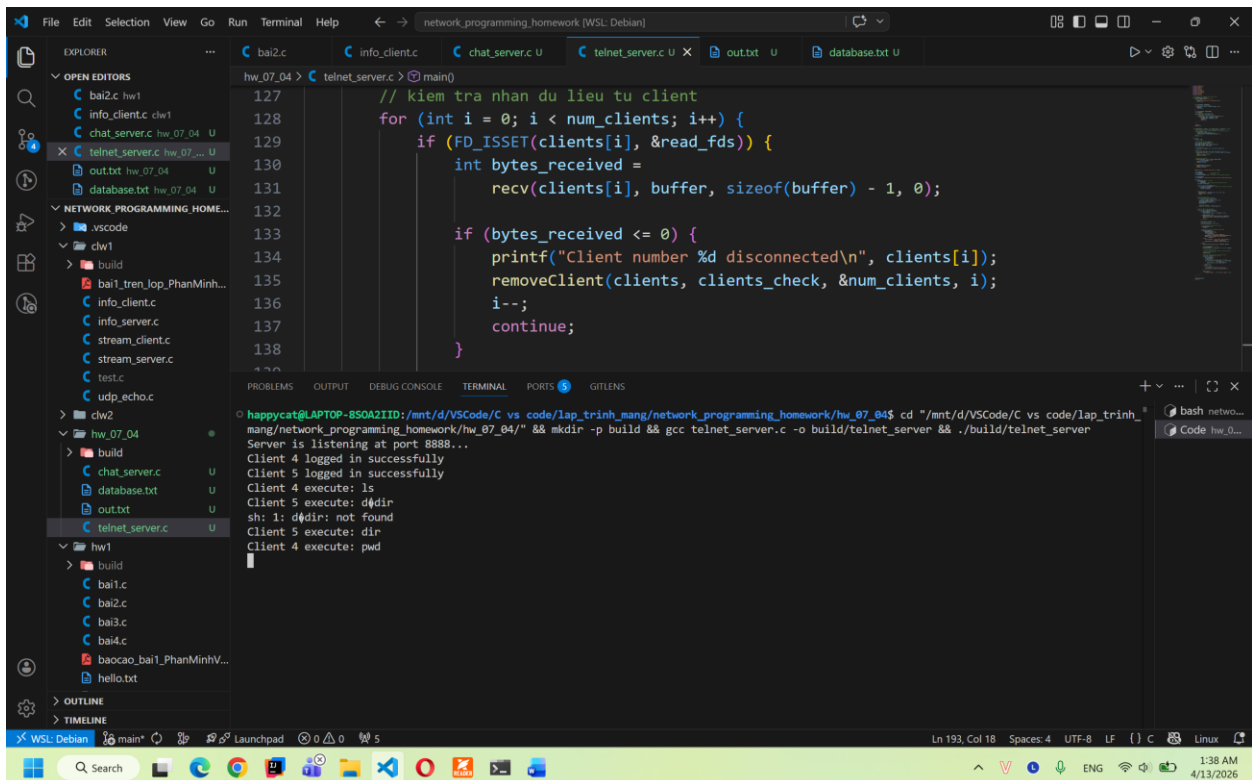
Client Terminal:

```
happycat@LAPTOP-850A2IID:/mnt/c/Users/Lenovo$ ncat 127.0.0.1 8888
Please enter login info (format: username password):
user user
Login successful. Enter command:
dir

Enter command: dir
build chat_server.c database.txt out.txt telnet_server.c

Enter command:
```

- Kết quả ở server



The image shows the VS Code interface with the 'telnet_server.c' file open. The code is in C and implements a simple telnet server. The terminal at the bottom shows the server's output, including the login process and the execution of commands from the client.

Code in telnet_server.c:

```
127 // kiểm tra nhận dữ liệu từ client
128 for (int i = 0; i < num_clients; i++) {
129     if (FD_ISSET(clients[i], &read_fds)) {
130         int bytes_received =
131             recv(clients[i], buffer, sizeof(buffer) - 1, 0);
132
133         if (bytes_received <= 0) {
134             printf("Client number %d disconnected\n", clients[i]);
135             removeClient(clients, clients_check, &num_clients, i);
136             i--;
137             continue;
138         }
139     }
140 }
```

Terminal Output:

```
happycat@LAPTOP-850A2IID:/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_07_04$ cd "/mnt/d/VSCode/C vs code/lap_trinh_mang/network_programming_homework/hw_07_04/" && mkdir -p build && gcc telnet_server.c -o build/telnet_server && ./build/telnet_server
Server is listening at port 8888...
Client 4 logged in successfully
Client 4 execute: ls
Client 5 execute: ddir
sh: 1: ddir: not found
Client 5 execute: dir
Client 4 execute: pwd
```